

PLAN OF ACTION — Galaxy Sports Edge iOS Application

Program: GSE Native (iOS-first) — Phase 6+ Item 3 from

`docs/product/phase-6-plus-planning.md` **Date opened:** 2026-09-15 **Executor:** Minis

(autonomous agent, iSH/Alpine aarch64 on iOS) **Owner:** Beexly (repo `Beexly/Sports`)

0. What already exists (reconnaissance findings — evidence, not assumption)

Decision on OPEN-P6-3 (reskin vs rebuild): Rebuild as a native client over the existing HTTP API. Rationale: the web app is Next.js 14 / React 18 with server components and Tailwind; shipping that inside a WebView would violate the app’s own doctrine (the design contract’s motion/telemetry rules, 44px targets, native haptics) and would not survive App Review as anything but a “repackaged website” (Guideline 4.2). The *design language* is reskinned (tokens ported 1:1); the *implementation* is native. The server stays the single source of truth — no business logic is duplicated client-side.

1. Non-negotiable constraints inherited from the repo

These are laws, not preferences. The app must not violate them.

1. **Rule 8 / positioning.** Never frame the engine as AI. Banned strings are machine-readable in `apps/web/lib/positioning-vocab.json`. The app ships its own copy of that list and a **runtime linter**.
 2. **No frontend-only paywalls (CLAUDE.md rule 3).** The app must never gate content that the server already sent it. Tier display is presentational; enforcement is the server’s. This inverts the usual mobile “unlock screen” pattern and is a deliberate architectural rule.
 3. **Confidence is not a probability.** Per `AGENTS.md` (2026-09-13): `confidence` is a weighted factor sum, measured anti-predictive at the top band, and rendering it as a percent reads as a win probability. The app renders it as a **score out of 100**, labelled “Confidence”, never `%`.
 4. **Never render a row the engine withheld.** `independentEdge.decision === "PASS"` and negative `expectedClv` rows are suppressed at source (`adverse-edge-suppression.ts`). The client mirrors the predicate as a **defensive** second layer and never re-orders by `confidence`.
 5. **Settlement badges are never colour-alone.** Monogram `W/L/P/V` per `DESIGN.md`.
 6. **No casino green.** `--verify` mint only, for settlement outcome.
 7. **Stale data never renders as fresh (rule 5).** Every data card carries a freshness stamp; a stale slate renders an honest “awaiting fresh data” state, not a cached board presented as live.
 8. **No emoji. No tout language.** Enforced by the same linter as rule 1.
 9. **Reduced motion honoured globally.** No animation on data values.
 10. **44×44pt minimum touch target.**
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2. App Store reality (this is the part most builds get wrong)

Research conducted and encoded into the app, not hand-waved:

Issue	Guideline	Resolution in this build
Selling Pro/Elite <i>inside the app</i> unlocks digital content	3.1.1 In-App Purchase	The app ships a StoreKit 2 subscription layer as the primary path (via <code>expo-iap</code> - equivalent module surface). Stripe remains the web path.
US external purchase links	US injunction (2025) permit external link-outs	Encoded as a region-gated “manage on web” affordance, default-off outside the US, never auto-opening a checkout.
Third-party sign-in (Google)	4.8 Sign in with Apple	Apple sign-in is implemented as a first-class provider, not an afterthought.
Account deletion	5.1.1(v)	In-app delete-account flow with server contract.
Gambling adjacency	5.3 / 1.4.3 , 17+ rating	No bet placement, no sportsbook deep links, no “bet now”, no odds-to-cash path. Age gate at first launch. Rating declared 17+ with the Gambling content descriptor, justified in the review notes.
Tracking	ATT	The app does no cross-app tracking and therefore does not request ATT. Declared in <code>PrivacyInfo.xcprivacy</code> .
Privacy manifest	Required since 2024	<code>PrivacyInfo.xcprivacy</code> with the exact required-reason API declarations for UserDefaults/FileTimestamp/DiskSpace.
Repackaged website	4.2 Minimum functionality	Native navigation, haptics, offline cache, Live Activity, push — substantively more than a wrapper.

3. Build plan (phased, each phase closes to done before the next opens)

P0 — Foundation

- `theme/` — FIELD tokens ported 1:1 from `design-tokens.css` (+ contrast metadata, confidence/risk ladders).
- `lib/voice.ts` — banned-phrase linter bound to the repo vocabulary file.
- `tsconfig/babel/metro/app.json/eas.json` — SDK 57, RN 0.86.3, React 19.2.3, New Architecture on.

P1 — Data core

- `api/client.ts` — timeout, retry w/ jitter, 429 `Retry-After` respect, 503 gate-body surfacing, ETag/If-None-Match, abort.
- `api/contracts.ts` — hand-written response types mirroring the server's real shapes (from the route source).
- `api/endpoints.ts` — one function per endpoint, all typed.
- `lib/freshness.ts`, `lib/entitlements.ts`, `lib/calibration.ts`, `lib/format.ts` — pure, unit-tested.

P2 — Offline + state

- `lib/cache.ts` (stale-while-revalidate, per-key TTL, honest staleness surfacing), `lib/queue.ts` (offline mutation queue), TanStack Query wiring.

P3 — Design system (native)

- Primitives: `Surface`, `EyeBrow`, `Numeral`, `Card`, `Divider`, `Pill`, `Button`, `GhostButton`.
- Signature components: `PickCard`, `ConfidenceMeter`, `EdgeBar`, `FactorTrail`, `SettlementBadge`, `FreshnessStamp`, `TierBadge`, `SourceTierBadge`, `CalibrationCurve` (SVG), `DiscriminationPanel`, `RiskDisclosure`, `EmptyState`, `HonestGate`.

P4 — Screens Board · Picks · Pick detail (+evidence drawer) · Calibration · Calibration/me · Brief · Performance · Watchlist · Game Room · Settings · Onboarding/age-gate · Paywall · Responsible play · Account deletion.

P5 — Platform integration

- Push (expo-notifications + server registration), Live Activity scaffold (expo-widgets), haptics, deep links, share, App Review prompt, background refresh.

P6 — Server contract (PR-ready, in-repo)

- `POST /api/mobile/v1/auth/exchange`, `POST /api/mobile/v1/devices`, `GET /api/mobile/v1/bootstrap`, `GET /api/mobile/v1/me`, `POST /api/mobile/v1/account/delete`, plus a bearer-auth middleware that reuses `getUserEntitlements`.

P7 — Store packaging

- `app.json` / `eas.json` / `store.config.json`, App Privacy answers, review notes, ASO copy, `PrivacyInfo.xcprivacy`, entitlements, export-compliance answer.

P8 — Verification

- `tsc --noEmit` clean against the **real** SDK 57 type surface; pure-logic test suite run under Node; design-preview HTML harness rendered + screenshotted for visual verification.

P9 — Review → improve → audit → polish

- Adversarial self-review, audit against doctrine + the 10 reference repos, then polish.

P10 — Iterate

- Round 2: 10 more repositories; repeat P9–P10 until the marginal return goes to zero.
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3b. Minis skills folded into this program

The Minis skill library is part of the toolchain for this build, not an aside. Explicit mapping:

Skill	Use in this program	Status
<code>asc-cli</code>	App Store Connect operations: install the <code>linux_arm64 asc</code> binary, stage the version, push metadata/localisations, TestFlight distribution, submit for review. Produces the exact owner-run command sequence.	Installed; awaiting the owner's <code>.p8</code> API key (cannot be created autonomously — Apple-account-bound).
<code>generative-ui-minis</code>	Visual verification harness. The app is React Native and this host has no simulator, so a generated HTML artifact renders the same tokens/components so screens can actually be <i>looked at</i> and iterated. Not a substitute for the app — a review instrument.	In use.
<code>nano-banana</code>	App icon / splash / marketing imagery generation.	Blocked — <code>GEMINI_API_KEY</code> unset. Fallback taken: the icon is drawn from the brand's own mark geometry (orbit + vector + core + ping) as SVG → PNG, which is more brand-true than a generated image anyway.
<code>pdf-converter</code>	Rendering App Review notes and the submission packet to PDF for the owner.	Queued.
<code>exa-search</code> / <code>web-search</code>	The reference-repository research rounds (P10) and App Store policy research.	In use.
<code>github-sync-helper</code>	Branch + PR mechanics for the in-repo server patch and any app-repo creation. Branch-and-PR by default, <code>never direct-to-main</code> .	Queued.
<code>s.apple-reminders</code> <code>nt</code> <code>whenpeak</code>	App Store release package (30% here, 70% there) in this's <code>butler</code> <code>gse-ios-release</code>	Queued (P10).

3c. Workstream P11 — X (Twitter) community, distribution and revenue

Requested by the owner, 2026-09-15. Not a side quest: it is the distribution half of the same product bet. Reconnaissance first, because the repo is further along here than expected.

What already exists (evidence)

Asset	Path	State
Voice spec — what the bot posts, refuses, and never says	docs/product/twitter-bot-voice-spec.md	Written, detailed, 8 hard refusals
Templates — 4 event kinds	apps/web/lib/twitter-bot/templates/{pick-publication,settlement,slate-state-gated,post-mortem-thread}.ts	Built
Planner — idempotency keys, blocked reasons, compliance gate	apps/web/lib/bot-outbox/plan.ts	Built
Adapters — DB record → template input	apps/web/lib/bot-outbox/records.ts	Built
Operator UI — review drafts before send	apps/web/app/cockpit/bot-outbox/page.tsx + preview route	Built
Eval cases — happy path, banned vocab, paid-pick refusal, loss	docs/ops/evals/twitter-bot-*.md	Written
Transport — auth, rate limit, send, retry, log	workers/twitter-bot/	ABSENT — this is the gap
Community / engagement / partnership layer	—	ABSENT — the owner's actual ask

So the honest finding is: **GSE has a fully specified, compliance-gated, operator-reviewable X bot that cannot post.** Everything up to the HTTP call exists. The workstream is therefore unusually well-defined.

Two doctrine conflicts found during reconnaissance (both must be fixed, not worked around)

1. **The spec renders confidence as a percent.** Its example is “*Published BOS –3.5 at 73% confidence*”. That spec predates the 2026-09-13 finding (`AGENTS.md`) that `confidence` is a weighted factor sum whose top band claims 0.8663 and realizes 0.5191 ($z = -10.7$). A percent on a public post asserts a win probability the number demonstrably is not — on the surface with the widest reach. The template must render `72/100` and say “score”, never “%”.
2. **The spec permits //.** `DESIGN.md` says settlement badges are monograms W/L/P/V and that “emoji $\approx$ zero”. These cannot both be right. Resolution: the *settlement glyph* is a documented exception on X (where a monogram is invisible at timeline scale and where the platform’s own convention carries meaning), but it may only appear on the settlement lead post and nowhere else. Every other post uses the sanctioned data glyphs. This gets written down and tested.

P11 deliverables

#	Deliverable	Why it is the right shape
1	<code>x-transport/</code> — OAuth 1.0a signer, API v2 client, rate limiter, idempotent sender, <code>MUTE_BOT</code> honouring, structured attempt log	The missing piece. Pure logic, unit-testable here (HMAC-SHA1 signing is deterministic).
2	Template corrections + tests for the two doctrine conflicts above	The templates are the voice; leaving them stale would ship the error.
3	<code>docs/product/x-community-strategy.md</code> — the community/engagement lane	The owner's ask. Growth mechanics that survive the no-engagement-bait rule.
4	Revenue/partnership framework — and an honest read on what is founder-gated	<code>docs/strategy/platform-gaps-triage.md</code> gates the affiliate/creator model on the founder. Say so.
5	Research lane — X as a <i>signal</i> source, wired through the existing <code>decision-genome/rumor-quarantine.ts</code>	X is a fast news surface. It is also a rumour surface. The repo already has the quarantine primitive; the lane must use it or not exist.
6	Eval files for the new surfaces, matching the spec's own required set	The spec names the evals; they should exist.

Sequencing: P11 runs after the P8/P9 verification and audit of the app, so the app is closed out before a second workstream opens. It is recorded here now so it is not lost.

3d. Cycle 1 status (2026-09-15) and the cycle 2 queue

Closed this cycle P0–P8 executed. Verification: 8/8 linter self-test · 41 files lint-clean · 105/105 logic tests · 18/18 OAuth tests (including the RFC 5849 vector) · full typecheck clean across all 42 files against the real Expo SDK 57 type surface (twice consecutively).

Phase	State	Evidence
P0 Foundation	Done	Tokens ported from the sole-authority file; voice linter bound to the repo vocabulary
P1 Data core	Done	27 client tests incl. gate bodies, 429 <code>Retry-After</code> , no silent cache
P2 Offline + state	Done	Labelled cache; <code>fromCache</code> never dressed as fresh
P3 Design system	Done	Primitives + 8 signature components
P4 Screens	Done	13 routes incl. age gate, paywall, account deletion, responsible play
P5 Platform integration	Partial	Push + notification routing done; Live Activity declared, target not written
P6 Server contract	Done	4 routes + bearer auth, PR-ready
P7 Store packaging	Done	<code>docs/store/SUBMISSION.md</code> , complete
P8 Verification	Done	Full typecheck clean (42 files); see the audit §2 for why it must still be re-run rather than trusted
P9 Review/audit	Done	13 defects found and fixed; 8 upstream findings
P10 Research round 1	Done	10 repositories, each tied to a decision it changed
P11 X community	Started	Strategy + verified OAuth signer; sender not written

Cycle 2 queue, in priority order

0. **Purchase reconciliation.** `app/paywall.tsx` finishes the StoreKit transaction before the server acknowledges, so a failed server call after a successful charge is unrecoverable: the customer paid and sees the free tier. Do not finish until acknowledged; set `appAccountToken`; reconcile on foreground. See `reports/01-review-and-audit.md` §5b and `research/round-02-repositories.md` §1. **This is above everything else, including the Mac build**, because it is the one item that loses money in production rather than in review.
1. **Build and run on a Mac** — `npm expo start --ios`, then `tsc --noEmit`. Nothing else is meaningful until the app has actually launched. Expect a handful of the class in audit §4 F7/F8.
2. **Component tests** — `@testing-library/react-native` over the four states that matter most: stale gate, bootstrap gate, redacted confidence, emptied board.
3. **X sender** — the signer is verified; what remains is the sender, the idempotency ledger, the scheduler wiring, and the two template-doctrine fixes.
4. **Wire `auth/exchange`** into the web sign-in callback (one line) and apply the Prisma migration.
5. **Live Activity target**, via `expo-apple-targets`.
6. **Research round 2** — StoreKit 2 state machines, offline sync conflict resolution, accessibility-first charting, privacy-preserving analytics, E2E on a simulator.

The definition-of-done clause that is not met, and cannot be met here

“It has been rendered and looked at.”

The app has never been rendered. The **harness** has — `preview/screens.html`, six frames, reviewed and fixed three times — and that is a mirror, not the app. This clause stays open until the first simulator run, and it is written here rather than quietly dropped because it is the difference between a design that has been seen and one that has only been described.

4. Definition of done for the whole program

A section is closed only when all of the following are true: 1. It typechecks. 2. Its pure logic has a passing test. 3. It has no banned vocabulary, no emoji, no casino green, no colour-alone signal. 4. It carries a freshness stamp if it shows data. 5. It has an honest empty/error/stale state. 6. It has been rendered and looked at. 7. It is listed in the audit ledger with an explicit remaining-gap note (or “none”).